1.Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

Ans.



2.List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Ans.

1. Modem

A modem is a networking device that connects our home or office network to the Internet Service Provider (ISP). It receives the internet signal and converts it into a form that can be used by networking devices.

Real-life example: A modem is used at home to get an internet connection from a broadband service provider.

2. Router

A router is used to connect different networks and share an internet connection with multiple devices. It sends data to the correct destination and many home routers also provide Wi-Fi connectivity.

Real-life example: A router is used at home to connect mobile phones, laptops, and smart TVs to the internet.

3. Switch

A switch is a networking device that connects multiple computers and other wired devices in a Local Area Network (LAN). It receives data and forwards it to the correct connected device.

Real-life example: A switch is used in a computer lab to connect many desktop computers through Ethernet cables.

4. Access Point (AP)

An access point is a device that allows wireless devices to connect to a wired network using Wi-Fi. It helps provide network coverage in areas where the existing Wi-Fi signal is weak or unavailable.

Real-life example: An access point is used in a college or office to provide Wi-Fi connectivity in different rooms.

5. Network Cable

A network cable is used to connect computers and networking devices so they can transfer data. Ethernet cables are commonly used for wired network connections and generally provide a stable and reliable connection.

3.Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

Ans.

1. Ethernet Cable (Twisted-Pair Cable)



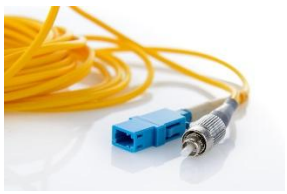
An Ethernet cable is commonly used to connect computers, routers, and switches in homes, offices, and computer laboratories. It provides a stable wired network connection.

2. Coaxial Cable



A coaxial cable has a central copper wire and protective shielding. It is commonly used for cable television, CCTV systems, and some broadband internet connections.

3. Fiber-Optic Cable



A fiber-optic cable transmits data using light signals. It is commonly used for high-speed internet connections and long-distance communication between buildings and cities.

4.Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Ans.

No.	Router	Switch
1.Main use	A router connects different networks and shares the internet connection.	A switch connects multiple devices within the same local network (LAN).
2.Typical location	It is commonly used in homes, offices, and colleges to provide internet access.	It is commonly used in computer labs, offices, and server rooms to connect computers.
3.How it handles data	It forwards data packets between networks using IP addresses.	It forwards data frames to the correct device using MAC addresses.
4.Internet connection	It connects a local network to the internet through an ISP connection.	It connects devices within a network; it does not normally provide internet access by itself.
5.Wireless facility	Many home routers provide Wi-Fi connectivity.	A normal switch does not provide Wi-Fi connectivity.
6.Example	A home router connects mobile phones and laptops to the internet.	A switch connects several desktop computers in a college computer lab.
7.Network layer	A router mainly works at the Network Layer (Layer 3) of the OSI model.	A standard switch mainly works at the Data Link Layer (Layer 2) of the OSI model.